

Gregory Gay

Chalmers and the University of Gothenburg
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Research Interests:

Software engineering, software testing, automated software engineering, use of AI in software engineering, AI engineering (SE applied to AI/ML-based systems), search-based software engineering, requirements engineering, data analytics, optimization, information retrieval.

Teaching Interests:

Software engineering, software testing, software verification & validation, software architecture, software design principles, artificial intelligence, optimization.

Education:

- Docent, University of Gothenburg, Gothenburg, Sweden, 2021.
- Ph.D. Computer Science, University of Minnesota, Minneapolis, Minnesota, 2015.
Advisor: Mats Heimdahl.
Thesis title: *Steering Model-Based Oracles to Admit Real Program Behaviors*.
- M.Sc. Computer Science, West Virginia University, Morgantown, West Virginia, 2010.
Advisor: Tim Menzies.
Thesis title: *Robust Optimization of Non-Linear Requirements Models*.
- B.Sc. Computer Science, West Virginia University, Morgantown, West Virginia, 2008.

Professional Experience:

- 2023–Present** Program Manager for Software Engineering and Technology M.Sc., Chalmers University of Technology, Gothenburg, Sweden.
- 2023–Present** Program Manager for Software Engineering and Management M.Sc., University of Gothenburg, Gothenburg, Sweden.
- 2021–Present** Associate Professor, Chalmers and the University of Gothenburg, Gothenburg, Sweden.
Interaction Design and Software Engineering Division,
Department of Computer Science & Engineering
- 2022** Program Manager (Temporary) for Software Engineering and Management B.Sc., University of Gothenburg, Gothenburg, Sweden.
- 2019–2021** Assistant Professor, Chalmers and the University of Gothenburg, Gothenburg, Sweden.
Interaction Design and Software Engineering Division,
Department of Computer Science & Engineering
- 2015–2019** Assistant Professor, University of South Carolina, Columbia, SC.
Department of Computer Science & Engineering
- 2010–2015** Research Assistant, University of Minnesota, Minneapolis, MN.
Critical Systems Group (under Mats Heimdahl)

- 2010** Visiting Researcher, Chinese Academy of Sciences, Beijing, PRC.
Lab for Internet Software Technologies, Institute of Software
- 2009** Intern, National Aeronautics and Space Administration (NASA), Mountain View, CA.
Robust Software Engineering Group, Ames Research Center
- 2007–2010** Research Assistant, West Virginia University, Morgantown, WV.
Modeling Intelligence Lab (under Tim Menzies)
- 2006–2007** Research Assistant, West Virginia University, Morgantown, WV.
Virtual Environments Lab (under Francis Van Scoy)
- 2005** SEAP Intern, National Aeronautics and Space Administration (NASA), Fairmont, WV.
Independent Verification & Validation Center

Teaching Experience:

- 2020–Present** Course Responsible, Software Quality and Testing (Bachelor, Chalmers and University of Gothenburg).
Course Materials (Newest): <https://greg4cr.github.io/courses/spring25dit636/index.html>
- 2023–Present** Course Responsible, Industrial Practice Project in Software Engineering (Master, Chalmers and University of Gothenburg).
- 2024–Present** Course Responsible, Research Project in Software Engineering (Master, Chalmers and University of Gothenburg).
- 2020–2022** Course Responsible, Software Engineering Principles for Complex Systems (Bachelor/Master, Chalmers and University of Gothenburg).
Course Materials (Newest): <https://greg4cr.github.io/courses/fall22tda594/index.html>
- 2019** Co-Teacher, Miniproject: Team Programming (Bachelor, University of Gothenburg).
Course Materials: <https://greg4cr.github.io/courses/misc/index.html>
- 2019** Co-Teacher, Mobile and Web Development (Bachelor, University of Gothenburg).
Course Materials: <https://greg4cr.github.io/courses/misc/index.html>
- 2019** Course Responsible, Software Engineering (Bachelor, University of South Carolina).
Course Materials: <https://greg4cr.github.io/courses/spring19csce247/index.html>
- 2018** Course Responsible, Software Architecture (Masters, University of South Carolina).
Course Materials: <https://greg4cr.github.io/courses/fall18csce742/index.html>
- 2016–2018** Course Responsible, Software Testing and Quality Assurance (Masters, University of South Carolina).
Course Materials (Newest): <https://greg4cr.github.io/courses/spring18csce747/index.html>
- 2016–2017** Course Responsible, Seminar on Advances in Computing (Masters, University of South Carolina).
Course Materials (Newest): <https://greg4cr.github.io/courses/fall17csce791/index.html>
- 2015–2017** Course Responsible, Software Engineering (Masters, University of South Carolina).
Course Materials (Newest): <https://greg4cr.github.io/courses/fall17csce740/index.html>
- 2015** Course Responsible, Software Engineering 1 (Bachelor/Master, University of Minnesota).
- 2012–2014** Teaching Assistant, Software Engineering 1 (Bachelor/Master, University of Minnesota).
- 2013** Teaching Assistant, Software Engineering 2 (Bachelor/Master, University of Minnesota).
- 2012** Participant, University of Minnesota Preparing Future Faculty Program.

Student Supervision:

Ph.D. Primary Supervisor (Ongoing and Graduated)

2025–Ongoing Yundong Shen, Chalmers University of Technology (estimated graduation: 2030).

2025–Ongoing Jingxiong Liu, Chalmers University of Technology (estimated graduation: 2030).

2016–2022 Alireza Salahirad, University of South Carolina.

2016–2020 Hussein Almulla, University of South Carolina.

Ph.D. Co-Supervisor (Ongoing and Graduated)

2025–Ongoing Lirong Yi, co-supervised with Philipp Leitner, Chalmers University of Technology (estimated graduation: 2030).

2020–2025 Khan Mohammad Habibullah, co-supervised with Jennifer Horkoff, University of Gothenburg.

Ph.D. Primary Supervisor (Incomplete)

2020–2025 Afonso Fontes, University of Gothenburg (chose to leave after completing licentiate).

2018–2019 Ying Meng, University of South Carolina (transferred to new supervisor after I left USC).

2018–2019 Peter Mourfield, University of South Carolina (transferred to new supervisor after I left USC).

2018–2019 Alaleh Torkjazi, University of South Carolina (transferred to new supervisor after I left USC).

M.Sc. Supervisor

2024 Linnea Wahlgren and Ludvig Lemner, Chalmers University of Technology.

2023 Viktor Tu and Albin Lönnfalt, Chalmers University of Technology.

2022–2023 (60 credit) Haozhao Lyu, Chalmers University of Technology.

2022 Rohini Bisht and Selomie Kindu Ejigu, Chalmers University of Technology and University of Gothenburg.

2022 Teklit Berihu Gereziher and Selam Gebrekrstos, Chalmers University of Technology.

2020 Ashish Husain and Martin Tran, Chalmers University of Technology.

2020 Rasmus Jenth, Chalmers University of Technology.

2018–2019 Burl Kenner III, University of South Carolina.

2017–2018 Srujana Bollina, University of South Carolina.

2016–2017 Ying Meng, University of South Carolina.

M.Sc. Co-Supervision

2026 Shahd Metwally and Elias Samantzis, co-supervised with Yundong Shen, University of Gothenburg.

2026 Zhuangzhuang Gong and Lialia Beniaminova, co-supervised with Jingxiong Liu, University of Gothenburg.

2026 Mohammadali Rashidfarokhi and Pradip Gotame, co-supervised with Francisco Gomes de Oliveira Neto, University of Gothenburg.

2025 Hampus Rhedin Stam and Anton Ekström, co-supervised with Francisco Gomes de Oliveira Neto, Chalmers University of Technology.

2025 Martin Engström and Jenny Lindevall, co-supervised with Mohannad Alhanahnah, Chalmers University of Technology.

2024 Avito Alexandre Costa da Silva, co-supervised with José Campos, University of Porto.

- 2024** Stefan Alexander van Heijningen and Theo Wiik, co-supervised with Francisco Gomes de Oliveira Neto, Chalmers University of Technology.
- 2024** Henrik Johansson and Erik Blomberg, co-supervised with Afonso Fontes, Chalmers University of Technology.
- 2022** Jonathan Örgård, co-supervised with Francisco Gomes de Oliveira Neto, University of Gothenburg.
- 2022** Lukas Berglund and Tim Grube, co-supervised with Francisco Gomes de Oliveira Neto, Chalmers University of Technology and University of Gothenburg.
- 2017–2018** George Akhvlediani, co-supervised with Duncan Buell, University of South Carolina.

B.Sc. Supervisor

- 2024** Arash Amiry, Nils Bengtsson Svanstedt, Anton Börås, Julia Giaro, and Elin Ruud, Chalmers University of Technology
- 2023** Gregory Sastrawidjaya and Edvin Danielsson, University of Gothenburg.
- 2023** Zubeen Maruf, University of Gothenburg.
- 2022** Shonaigh Douglas, University of Gothenburg.
- 2022** Dia Istanbuly and Max Zimmer, University of Gothenburg.
- 2020** Fabian Daneshmand-Mehr and Daniel Salomons, University of Gothenburg.
- 2020** Sarkis George Sarkisian, University of Gothenburg.

B.Sc. Co-Supervision

- 2025** Adrian Hassa, Ionel Alejandro Pop Jara, and Teodora Portase, co-supervised with Francisco Gomes de Oliveira Neto, University of Gothenburg.
- 2023** Karl Stahre and Malte Götharsson, co-supervised with Francisco Gomes de Oliveira Neto, University of Gothenburg.

Independent Study, Research Project, or Research Internship Supervisor¹

- 2025** Shun Ikumi, University of Tokyo (Research Project).
- 2025** Shahd Metwally, University of Gothenburg (Research Assistant).
- 2024** Nicole Andrea Quinstedt, University of Gothenburg (Research Assistant).
- 2024** Juan Garcia Diaz, University of Gothenburg (Research Assistant, Co-supervised with Jennifer Horkoff).
- 2023** Albin Landgren, Chalmers University of Technology (Industrial Practice Project).
- 2018** Hayley Lichtenfels, University of South Carolina (Independent Study).
- 2016** Allen Kanapala, University of South Carolina (Independent Study).
- 2016** Narasimha Chilukuri, University of South Carolina (Independent Study).
- 2016** Craig Sharp, University of South Carolina (Independent Study).

Ph.D. Examination/Opponent

- Expected in 2026** Alaleh Torkjazi, Ph.D. Examination Committee, University of South Carolina.
- Expected in 2026** Amid Golmohammadi, External Ph.D. Examiner, Kristiania University of Applied Sciences
- 2026** Clinton Shen Cao, Ph.D. Examination Committee, TU Delft.
- 2026** Stephan Lukasczyk, External Ph.D. Examiner, University of Passau.

¹An “Independent Study” or an “Industrial Practice/Research Project” is similar to a Bachelor’s Thesis in scope.

2025 Islam Elgendy, External Ph.D. Examiner, University of Sheffield.
2025 Waleed Abdeen, Ph.D. Examination Committee, Blekinge Institute of Technology.
2024 Stefan Karlsson, Ph.D. Examination Committee, Mälardalen University.
2024 Mitchell Olsthoorn, Ph.D. Examination Committee, TU Delft.
2023 Milos Ojdanic, Ph.D. Examination Committee, University of Luxembourg.
2022 William Hoskins, Ph.D. Examination Committee, University of South Carolina.
2022 Anjana Perera, External Ph.D. Examiner, Monash University.
2021 Khouloud Gaaloul, Ph.D. Examination Committee, University of Luxembourg.

Licentiate Examination/Opponent

2023 Andreas Bauer, Licentiate Opponent, Blekinge Institute of Technology.
2022 Qunying Song, Licentiate Opponent, Lund University.
2020 Mahshid Helali, Licentiate Opponent, Mälardalen University.

M.Sc. Examination

2020–Present Examined 39 theses, Chalmers and University of Gothenburg.
2026 Joo Pedro Pedrosa Domingues, Universidade da Beira Interior.
2024 Manuel Ferreira Magalhaes, Universidade da Beira Interior.

B.Sc. Examination

2020–Present Examined 10 theses, University of Gothenburg.

Funding:

2027 University of Gothenburg Faculty of Science and Technology, Sabbatical and Network Program (Sole PI, 768,227 SEK)
2025–Present Wallenberg Autonomous Systems, AI, and Software Program (WASP), TestExtender: Using Large Language Models to Augment Software Test Suites (Sole PI, 3,600,000 SEK)
2025–Present Wallenberg Autonomous Systems, AI, and Software Program (WASP), Exploring the Integration of Large Language Models in Industrial Test Maintenance Processes (Sole PI, 3,600,000 SEK)
2025–Present Software Center, Trustworthy and Responsible AI-Centric Test Engineering (Co-PI, 10% of salary funded on an ongoing basis).
2024–2025 Software Center, AI-Enabled Test Automation, Generation, and Optimization (Co-PI, 10% of salary during funding period).
2024–2025 Software Center, Trustworthy and Human-Centered Test Automation (Co-PI, 10% of salary during funding period).
2020–2025 Vetenskapsrådet (Swedish Research Council) Award 2019-05275, Context-Infused Automated Software Test Generation (Sole PI, 3,900,000 SEK).
2020–2024 Software Center, Aspects of Automated Testing (Co-PI, 15–20% of salary during funding period).
2019–2020 South Carolina NASA EPSCoR, Robust Software Testing of Autonomous Aerospace Robotic Systems Using Transfer Learning (Co-PI, \$25,000.00).
2018–2019 University of South Carolina ASPIRE-1, Investigating the Relationship between Real and Synthetic Software Faults (Sole PI, \$14,959.00).
2017–2019 National Science Foundation Award CCF-1657299, CRII: SHF: Understanding The Role of Software Test Adequacy Criteria in Search-Based Test Generation (Sole PI, \$173,528.00).

Awards:

- 2024** Distinguished Reviewer, ACM Transactions on Software Engineering and Methodology.
- 2023** Best Paper Award, 35th International Conference on Testing Software and Systems (ICTSS'23), *Understanding Problem Solving in Software Testing: An Exploration of Tester Routines and Behavior*.
- 2023** Distinguished Reviewer, 27th International Conference on Evaluation and Assessment in Software Engineering (EASE'23).
- 2023** Distinguished Paper Award, 16th International Conference on Software Testing, Verification, and Validation (ICST'23), *How Closely are Common Mutation Operators Coupled to Real Faults?*.
- 2020** Distinguished Reviewer, 35th International Conference on Automated Software Engineering (ASE'20).
- 2019** 2009-2019 Most Influential Paper Award, 35th International Conference on Software Maintenance and Evolution (ICSME'19), *On the Use of Relevance Feedback in IR-based Concept Location*.
- 2019** Best Reviewer, Journal of Systems and Software.
- 2018** Graduate Teaching Award, University of South Carolina (Department of Computer Science & Engineering).
- 2018** Challenge Award Winner, 10th Symposium on Search-Based Software Engineering (SSBSE'18), *Detecting Real Faults in the Gson Library Through Search-Based Unit Test Generation*.
- 2018** Best Presentation, 11th International Workshop on Search-Based Software Testing (SBST'18), *Multifaceted Test Suite Generation Using Primary and Supporting Fitness Functions*.
- 2016** Challenge Award Winner, 8th Symposium on Search-Based Software Engineering (SSBSE'16), *Challenges in Using Search-Based Test Generation to Identify Real Faults in Mockito*.
- 2014** Best Presentation, 7th International Workshop on Search-Based Software Testing (SBST'14), *Moving the Goalposts: Coverage Satisfaction is Not Enough*.
- 2010–2013** National Science Foundation Graduate Research Fellowship.

Conference Steering Committees and Chairmanships:

- 2025–Present** Steering Committee Co-Chair, International Conference on Software Testing, Verification, and Validation (ICST)
- 2023–Present** Steering Committee, International Conference on Software Testing, Verification, and Validation (ICST)
- 2025–Present** Co-Chair, International Workshop on evaluation and assessment in software Engineering Education and Training (LEARNER)
- 2027** Sponsorship Chair, 18th International Conference on Performance Engineering (ICPE'27).
- 2027** Workshop Co-Chair, 20th International Conference on Software Testing, Verification, and Validation (ICST'27).
- 2026** Review Process Co-Chair, 41st IEEE/ACM International Conference on Automated Software Engineering (ASE'26)
- 2026** Hot-off-the-Press Track Co-Chair, 18th Symposium on Search-Based Software Engineering (SSBSE'26).
- 2024** Program Co-Chair, 17th International Conference on Software Testing, Verification, and Validation (ICST'24)
- 2023–2024** Co-Chair, International Workshop on Artificial Intelligence in Software Testing (AIST)
- 2023** New Ideas and Emerging Results/Replications and Negative Results (NIER/RENE) Track Co-Chair, 15th Symposium on Search-Based Software Engineering (SSBSE'23).

2023 Testing Tools/Tool Demonstrations Co-Chair, 16th International Conference on Software Testing, Verification, and Validation (ICST'23)

2016–2022 Steering Committee, Symposium on Search-Based Software Engineering (SSBSE).

2022 Publicity and Social Media Co-Chair, 38th International Conference on Software Maintenance and Evolution (ICSME'22)

2021 Artifact Evaluation Co-Chair, 37th International Conference on Software Maintenance and Evolution (ICSME'21)

2021 Challenge Track Co-Chair, 13th Symposium on Search-Based Software Engineering (SSBSE'21).

2020 New Ideas and Emerging Results (NIER) Track Co-Chair, 12th Symposium on Search-Based Software Engineering (SSBSE'20).

2015–2020 Steering Committee, International Workshop on Search-Based Software Testing (SBST).

2019 Program Co-Chair, 11th Symposium on Search-Based Software Engineering (SSBSE'19).

2019 Co-Chair, 3rd ROSE Festival (Recognizing and Rewarding Open Science in Software Engineering, ESEC/FSE Special Track).

2019 General Chair, 6th International Workshop on Requirements Engineering and Testing (RET'19)

2018 Workshop Co-Chair, 11th International Conference on Software Testing, Verification, and Validation (ICST'18).

2018 Program Co-Chair, 5th International Workshop on Requirements Engineering and Testing (RET'18)

2017–2018 Steering Committee Deputy Chair, International Workshop on Search-Based Software Testing (SBST).

2017 Publicity Co-Chair, 9th Symposium on Search-Based Software Engineering (SSBSE'17).

2017 Co-Chair, 4th International Workshop on Requirements Engineering and Testing (RET'17).

2016–2017 Steering Committee Chair, International Workshop on Search-Based Software Testing (SBST).

2016 Co-Chair, 9th International Workshop on Search-Based Software Testing (SBST'16).

2016 Program Co-Chair, 3rd International Workshop on Requirements Engineering and Testing (RET'16).

2015 Co-Chair, 8th International Workshop on Search-Based Software Testing (SBST'15).

2012 North America Publicity Chair, 27th IEEE/ACM International Conference on Automated Software Engineering (ASE'12).

2012 Web Chair, 20th IEEE International Conference on Requirements Engineering (RE'12).

2008–2010 Web Chair, International Conference on Predictive Models in Software Engineering (PROMISE).

Conference Program Committees and Journal Reviewing:

2024–Present Program Committee, ACM/IEEE International Conference on Automation of Software Test (AST)

2023–Present Program Committee, International Conference on Evaluation and Assessment in Software Engineering (EASE).

2018–Present Program Committee, Genetic and Evolutionary Computation Conference (GECCO)

2018–Present Reviewer, Information and Software Technology

2017–Present Program Committee, International Conference on Advances in System Testing and Validation Lifecycle (VALID).

2017–Present Program Committee, International Workshop on Search-Based Software Testing (SBST).

2016–2018, 2023, 2025–Present Program Committee, Symposium on Search-Based Software Engineering (SSBSE).

2016–Present Reviewer, Journal of Systems and Software

2015–Present Reviewer, Empirical Software Engineering Journal

2015–Present Reviewer, ACM Transactions on Software Engineering and Methodology

2014–Present Reviewer, IEEE Transactions on Software Engineering

2026 Program Committee, International Symposium on Software Testing and Analysis (ISSTA)

2018–2019, 2026 Program Committee, International Conference on Software Testing, Verification, and Validation (ICST)

2022, 2025 Program Committee, International Conference on Software Engineering (ICSE).

2020, 2022, 2025 Program Committee, Symposium on Search-Based Software Engineering (SSBSE) (Challenge Track)

2025 Program Committee, International Workshop on Mutation Testing (Mutation).

2025 Program Committee, IFIP International Conference on Testing Software and Systems (ICTSS).

2025 Program Committee, Symposium on Search-Based Software Engineering (SSBSE) (Hot off the Press Track)

2020, 2022–2024 Program Committee, Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE) (Tool Demonstrations)

2019, 2022–2024 Program Committee, International Symposium on Software Testing and Analysis (ISSTA) (Tool Demonstrations)

2020–2024 Program Committee, Symposium on Search-Based Software Engineering (SSBSE) (Replications and Negative Results Track)

2023 Program Committee, Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE) (Student Research Competition)

2020–2022 Program Committee, International Conference on Automated Software Engineering (ASE)

2019–2022 Program Committee, International Conference on Automated Software Engineering (ASE) (Tool Demonstrations)

2023 Program Committee, Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE) (Student Research Competition)

2022 Program Committee, International Workshop on Artificial Intelligence in Software Testing (AIST).

2010, 2021 Reviewer, Automated Software Engineering (journal)

2021 Program Committee, International Conference on the Applications of Evolutionary Computing (EvoApplications)

2021 Program Committee, International Conference on Software Analysis, Evolution and Reengineering (SANER) (Tool Track)

2021 Program Committee, International Workshop on Test Oracles (TORACLE)

2019–2020 Reviewer, IEEE Access

2017–2020 Reviewer, IET Software

2014, 2018, 2020 Reviewer, Software Testing, Verification and Reliability

2020 Program Committee, International Conference on Software Engineering (ICSE) (Poster Track).

2020 Reviewer, Software and Systems Modeling

2018–2019 Program Committee, International Conference on Software Engineering (ICSE) (Demonstrations Track).

2017–2019 Program Committee, European Conference on the Applications of Evolutionary Computing (EvoSET Track—Nature-inspired algorithms in Software Engineering and Testing).

2019 Program Committee, International Workshop on Software Engineering Intelligence (SEI)
2019 Reviewer, Software Practice and Experience
2019 Reviewer, Systems Engineering
2017–2018 Program Committee, International Workshop on Software Analytics (SWAN).
2012, 2018 Reviewer, Software Quality Journal
2018 Reviewer, Journal of Software: Evolution and Process
2018 Reviewer, IEEE Transactions on Reliability
2018 Reviewer, Applied Soft Computing Journal
2018 Reviewer, Traffic Injury Prevention
2016–2017 Reviewer, IEEE Transactions on Evolutionary Computation
2017 Reviewer, The Computer Journal
2017 Reviewer, Formal Methods in System Design
2016 Reviewer, Journal of Classification
2016 Reviewer, 2017 IFAC World Conference
2015 Program Committee, International Workshop on Actionable Analytics for SE (ACTION).
2014 Reviewer, Journal of Aerospace Information Systems
2013 Reviewer, IEEE Software
2012 Reviewer, Formal Methods for Industrial Critical System
2011–2012 Program Committee, International Conference on Predictive Models in Software Engineering (PROMISE).

Additional Professional Activities:

2026–Present Associate Editor, IEEE Transactions on Software Engineering
2024–Present Deputy Editor-in-Chief, Automated Software Engineering Journal
2024–Present Cluster Co-Chair (Software Engineering and Technology), Wallenberg Autonomous Systems, AI, and Software Program (WASP)
2023–Present Member, ACM TOSEM Replicated Computational Results Distinguished Reviewers Board
2019–Present Member, ACM TOSEM Board of Distinguished Reviewers
2017–Present Member, Empirical Software Engineering Journal Review Board
2027 Award Committee, IEEE Technical Council on Software Engineering
2026 Invited Participant, Shonan Seminar 261: Green Intelligence: Sustainable Development and Operations of Intelligent Systems
2026 Reviewer, WASP Academic Ph.D. Funding Call (Software Track)
2024 Award Committee, ACM SIGSOFT
2023 Reviewer, CHIST-ERA Funding Call Open & Re-usable Research Data & Software
2023 Panelist, ICST 2023 Doctoral Symposium
2023 Invited Participant, Dagstuhl Seminar 23103: Testing and Debugging of Data Analysis Workflows
2020 Panelist, ICSE Student Mentoring Workshop
2019 Invited Participant, Shonan Seminar 160: Fuzzing and Symbolic Execution: Reflections, Challenges, and Opportunities
2018 Panelist, NSF Panel P181594 (CRI-SW)

2013 Student Volunteer, International Conference on Software Engineering
2012 Student Volunteer, 2012 International Symposium on Software Testing and Analysis
2008–2009 President, ACM West Virginia University Student Chapter
2007–2008 Vice-President, ACM West Virginia University Student Chapter
2007–2010 Member, West Virginia University Engineering Student Advisory Council

Published Books and Book Chapters:

Names in **bold** are supervised students.

1. **Afonso Fontes**, Gregory Gay, Francisco Gomes de Oliveria Neto, Robert Feldt. Automated Support for Unit Test Generation. Book chapter, *Optimising the Software Development Process with Artificial Intelligence*. Springer, 2023. ISBN: 978-981-19-9948-2. Available at <http://greg4cr.github.io/pdf/21ai4se.pdf>.
2. Shiva Nejati and Gregory Gay, editors. Proceedings of the 11th International Symposium on Search-Based Software Engineering, SSBSE 2019, Tallinn, Estonia, August 31-September 1, 2019. Vol. 11664. Lecture Notes in Computer Science. Springer, 2019. ISBN: 978-3-030-27455-9.

Journal Publications:

3. **Jingxiong Liu**, **Ludvig Lemner**, **Linnea Wahlgren**, Gregory Gay, Nasser Mohammadiha, Joakim Wennerberg. Exploring the Integration of Large Language Models in Industrial Test Maintenance Processes. *Journal of Systems and Software*. To appear, 2026. Available at <http://greg4cr.github.io/pdf/26maintenancellm.pdf>.
4. **Afonso Fontes**, Gregory Gay, Robert Feldt. Exploring the Interaction of Code Coverage and Non-Coverage Objectives in Search-Based Test Generation. *Wiley Software Testing, Verification and Reliability*. To appear, 2025. Available at <http://greg4cr.github.io/pdf/25multiobj.pdf>.
5. **Albin Lönnfalt**, **Viktor Tu**, Gregory Gay, Animesh Singh, Sahar Tahvili. An Intelligent Test Management System for Optimizing Decision Making During Software Testing. *Journal of Systems and Software*. Volume 219. 2024. Available at <http://greg4cr.github.io/pdf/24CIConfigPrediction.pdf>.
6. **Khan Mohammad Habibullah**, Hans-Martin Heyn, Gregory Gay, Jennifer Horkoff, Eric Knauss, Markus Borg, Alessia Knauss, Håkan Sivencrona, Polly Jing Li. Requirements and Software Engineering for Automotive Perception Systems: An Interview Study. *Requirements Engineering Journal*. Volume 29, Number 1. March 2024. Available at <http://greg4cr.github.io/pdf/24aps.pdf>.
7. **Afonso Fontes**, Gregory Gay. The Integration of Machine Learning into Automated Test Generation: A Systematic Mapping Study. *Wiley Software Testing, Verification and Reliability*. Volume 33, Issue 4. 2023. Available at <http://greg4cr.github.io/pdf/23mapping.pdf>.
8. **Khan Mohammad Habibullah**, Gregory Gay, Jennifer Horkoff. Non-Functional Requirements for Machine Learning: Understanding Current Use and Challenges Among Practitioners. *Requirements Engineering Journal*. Volume 28, Number 2. June 2023. Available at <http://greg4cr.github.io/pdf/23nfrsurvey.pdf>.
9. **Alireza Salahirad**, Gregory Gay, Ehsan Mohammadi. Mapping the Structure and Evolution of Software Testing Research Over the Past Three Decades. *Journal of Systems and Software*. Volume 195. 2023. Available at <http://greg4cr.github.io/pdf/22topics.pdf>.
10. **Hussein Almulla**, Gregory Gay. Learning How to Search: Generating Effective Test Cases Through Adaptive Fitness Function Selection. *Empirical Software Engineering Journal*. Volume 27, Article 38. 2022. Available at <http://greg4cr.github.io/pdf/21affs.pdf>.

11. **Alireza Salahirad, Hussein Almulla**, Gregory Gay. Choosing The Fitness Function for the Job: Automated Generation of Test Suites that Detect Real Faults. *Wiley Software Testing, Verification and Reliability*. Volume 29, Issue 4-5. June–August 2019. Available at <http://greg4cr.github.io/pdf/19fitness.pdf>.
12. **Ying Meng**, Gregory Gay, Michael Whalen. Ensuring the Observability of Structural Test Obligations. *IEEE Transactions on Software Engineering*. Volume 46, Issue 7, September 2018. Available at <http://greg4cr.github.io/pdf/18omcdc.pdf>.
13. Amanda Schwartz, Daniel Puckett, **Ying Meng**, Gregory Gay. Investigating Faults Missed by Test Suites Achieving High Code Coverage. *Journal of Systems and Software*. Volume 144. October 2018. Pages 106-120. Available at <http://greg4cr.github.io/pdf/18mutation.pdf>.
14. Gregory Gay, Sanjai Rayadurgam, Mats P.E. Heimdahl. Automated Steering of Model-Based Test Oracles to Admit Real Program Behaviors. *IEEE Transactions on Software Engineering*. Volume 43, Number 6. June 2017. Pages 531-555. Available at <http://greg4cr.github.io/pdf/16steering.pdf>.
15. Gregory Gay, Ajitha Rajan, Matt Staats, Michael Whalen, Mats P.E. Heimdahl. The Effect of Program and Model Structure on the Effectiveness of MC/DC Test Adequacy Coverage. *ACM Transactions on Software Engineering and Methodology*. Volume 25, Number 3, Article 25. August 2016. Available at <http://greg4cr.github.io/pdf/16mcdc.pdf>.
16. Gregory Gay, Matt Staats, Michael Whalen, Mats P.E. Heimdahl. Automated Oracle Data Selection Support. *IEEE Transactions on Software Engineering*. Volume 41, Number 11. November 2015. Pages 1119-1137. Available at <http://greg4cr.github.io/pdf/15oracles.pdf>.
17. Gregory Gay, Matt Staats, Michael Whalen, Mats P.E. Heimdahl. The Risks of Coverage-Directed Test Case Generation. *IEEE Transactions on Software Engineering*. Volume 41, Number 8. August 2015. Pages 803-819. Available at <http://greg4cr.github.io/pdf/15covrisks.pdf>.
18. Adam Nelson, Tim Menzies, Gregory Gay. Sharing Experiments Using Open-Source Software. *Software: Practice and Experience*. Volume 41, Number 3. March 2011. Pages 283-305. Available at <http://greg4cr.github.io/pdf/10ourmine.pdf>.
19. Gregory Gay, Tim Menzies, Misty Davies, and Karen Gundy-Burlet. Automatically Finding the Control Variables for Complex System Behavior. *Automated Software Engineering*. Volume 17, Number 4. December 2010. Pages 1-30. Available at <http://www.greg4cr.github.io/pdf/10tar3.pdf>.
20. Gregory Gay, Tim Menzies, Omid Jalali, Gregory Mundy, Beau Gilkerson, Martin Feather, and James Kiper. Finding Robust Solutions in Requirements Models. *Automated Software Engineering*. Volume 17, Number 1. March 2010. Pages 87-116. Available at <http://www.greg4cr.github.io/pdf/10keys.pdf>.

Conference Publications:

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22. **Anton Ekström, Hampus Rhedin Stam**, Francisco Gomes de Oliveira Neto, Gregory Gay, Sabina Edenlund. From Logs to Lessons: An Exploration of LLM-based Log Summarization for Debugging Automotive Software. *Proceedings of the 7th ACM/IEEE International Conference on Automation of Software Test (AST'26)*. Rio de Janeiro, Brazil, April 2026. Available at <http://greg4cr.github.io/pdf/26llmlog.pdf>.
23. **Khan Mohammad Habibullah, Juan Garcia Diaz**, Gregory Gay, Jennifer Horkoff. Maintainability Definition, Scoping, and Measurement for Machine Learning Systems. *Proceedings of the 18th International Conference on the Quality of Information and Communications Technology*

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24. **Shun Ikumi**, Gregory Gay, Maiko Sakamoto. Discourse Analysis for Participatory Discussions: A Topic Shift Detection Approach Using Text Mining and Network-Based Analysis. *Proceedings of the 25th International Conference on Group Decision and Negotiation, Doctoral Consortium Track (GDN'25)*. Zaragoza, Spain, June 2025. Available at <http://greg4cr.github.io/pdf/25gdn.pdf>.
 25. **Stefan Alexander van Heijningen, Theo Wiik**, Francisco Gomes de Oliveira Neto, Gregory Gay, Kim Viggedal, David Friberg. Integrating Mutation Testing Into Developer Workflow: An Industrial Case Study. *Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering, Industry Track (ASE'24)*. Sacramento, USA, October 2024. Available at <http://greg4cr.github.io/pdf/24mutation.pdf>.
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 30. **Haozhou Lyu**, Gregory Gay, Maiko Sakamoto. Developer Views on Software Carbon Footprint and its Potential for Automated Reduction. *Proceedings of 15th Symposium on Search-Based Software Engineering (SSBSE'23)*. San Francisco, USA, December 2023. Available at <http://greg4cr.github.io/pdf/23cfreq.pdf>.
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38. Hamid Ebadi, Mahshid Helali Moghadam, Markus Borg, Gregory Gay, **Afonso Fontes**, Kasper Socha. Efficient and Effective Generation of Test Cases for Pedestrian Detection–Search-based Software Testing of Baidu Apollo in SVL. *Proceedings of 3rd IEEE International Conference on Artificial Intelligence Testing, Challenge Track (AITest'21)*. Bari, Italy, August 2021. Available at <http://greg4cr.github.io/pdf/21aitest.pdf>.
39. Gregory Gay, René Just. Defects4J as a Challenge Case for the Search-Based Software Engineering Community. *Proceedings of 12th Symposium on Search-Based Software Engineering, Challenge Cases Track (SSBSE'20)*. Bari, Italy, September 2020. Available at <http://greg4cr.github.io/pdf/20d4j.pdf>.
40. **Hussein Almulla**, Gregory Gay. Generating Diverse Test Suites for Gson Through Adaptive Fitness Function Selection. *Proceedings of 12th Symposium on Search-Based Software Engineering, Challenge Solutions Track (SSBSE'20)*. Bari, Italy, September 2020. Available at <http://greg4cr.github.io/pdf/20rldiv.pdf>.
41. **Srujana Bollina**, Gregory Gay. Bytecode-based Multiple Condition Coverage: An Initial Investigation. *Proceedings of 12th Symposium on Search-Based Software Engineering, Replications and Negative Results Track (SSBSE'20)*. Bari, Italy, September 2020. Available at <http://greg4cr.github.io/pdf/20bmcc.pdf>.
42. **Ying Meng**, Gregory Gay. Understanding The Impact of Solver Choice in Model-Based Test Generation. *Proceedings of the ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM'20)*. Bari, Italy, September 2020. Available at <http://greg4cr.github.io/pdf/20solvers.pdf>.
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46. **Hussein Almulla**, **Alireza Salahirad**, Gregory Gay. Using Search-Based Test Generation to Discover Real Faults in Guava. *Proceedings of the 9th Symposium on Search-Based Software Engi-*

- neering, *Challenge Track (SSBSE'17)*. Paderborn, Germany, September 2017. Available at <http://greg4cr.github.io/pdf/17guava.pdf>.
47. Gregory Gay. Generating Effective Test Suites by Combining Coverage Criteria. *Proceedings of the 9th Symposium on Search-Based Software Engineering (SSBSE'17)*. Paderborn, Germany, September 2017. Available at <http://greg4cr.github.io/pdf/17ssbse.pdf>.
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 49. Gregory Gay. Challenges in Using Search-Based Test Generation to Identify Real Faults in Mockito. *Proceedings of the 8th Symposium on Search-Based Software Engineering, Challenge Track (SSBSE'16)*. Raleigh, NC, USA, October 2016. **Best Paper Winner (Challenge Track)**. Available at <http://greg4cr.github.io/pdf/16mockito.pdf>.
 50. Dongjiang You, Sanjai Rayadurgam, Michael Whalen, Mats P.E. Heimdahl, Gregory Gay. Efficient Observability-based Test Generation by Dynamic Symbolic Execution. *Proceedings of the 26th IEEE International Symposium on Software Reliability Engineering (ISSRE'15)*. Gaithersburg, MD, USA, November 2015. Available at <http://greg4cr.github.io/pdf/15issre.pdf>.
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 53. Michael Whalen, Gregory Gay, Dongjiang You, and Mats P.E. Heimdahl. Observable Modified Condition/Decision Coverage. *Proceedings of the 35th ACM/IEEE International Conference on Software Engineering (ICSE'13)*. San Francisco, United States, May 2013. Available at <http://greg4cr.github.io/pdf/13omcdc.pdf>.
 54. Matt Staats, Gregory Gay, and Mats P.E. Heimdahl. Automated Oracle Creation Support, or: How I Learned to Stop Worrying About Fault Propagation and Love Mutation Testing. *Proceedings of the 34th ACM/IEEE International Conference on Software Engineering (ICSE'12)*. Zurich, Switzerland, May 2012. Available at <http://greg4cr.github.io/pdf/12oracle.pdf>.
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 56. Ekrem Kocaguneli, Gregory Gay, Tim Menzies, Ye Yang, and Jacky Keung. When to Use Data from Other Projects for Effort Estimation. Short Paper, *Proceedings of the 25th ACM/IEEE International Conference on Automated Software Engineering (ASE'10)*. Antwerp, Belgium, September 2010. Available at <http://greg4cr.github.io/pdf/10ccwc.pdf>.
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 58. Jia Chen, Ye Yang, Wen Zhang, Gregory Gay. Measuring the Heterogeneity of Crosscompany Datasets. *Proceedings of the 11th International Conference on Product Focused Software Development and Process Improvement (PROFES'10)*. Limerick, Ireland, June 2010. Available at <http://greg4cr.github.io/pdf/10profes.pdf>.

59. Gregory Gay, Sonia Haiduc, Andrian Marcus, Tim Menzies. On the Use of Relevance Feedback in IR-based Concept Location. *Proceedings of the 25th IEEE International Conference on Software Maintenance (ICSM'09)*. Alberta, Canada, September 2009. **2009–2019 Most Influential Paper Award**. Available at <http://greg4cr.github.io/pdf/09irrf.pdf>.
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Workshop Publications:

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62. **Jonathan Örgård**, Gregory Gay, Francisco Gomes de Oliveira Neto, Kim Viggedal. Mutation Testing in Continuous Integration: An Exploratory Industrial Case Study. *Proceedings of the 18th International Workshop on Mutation Analysis (MUTATION'23)*. Dublin, Ireland, April 2023. Available at <http://greg4cr.github.io/pdf/23mutationci.pdf>.
63. **Khan Mohammad Habibullah**, Gregory Gay, Jennifer Horkoff. Non-Functional Requirements for Machine Learning: An Exploration of System Scope and Interest. *Proceedings of the 1st Workshop on Software Engineering for Responsible AI (SE4RAI'22)*. Pittsburgh, USA, May 2022. Available at <http://greg4cr.github.io/pdf/22nfrexplore.pdf>.
64. **Afonso Fontes**, Gregory Gay. Using Machine Learning to Generate Test Oracles: A Systematic Literature Review. *Proceedings of the 1st International Workshop on Test Oracles (TORACLE'21)*. Athens, Greece, August 2021. Available at <http://greg4cr.github.io/pdf/21oracleslr.pdf>.
65. Gregory Gay. One-Size-Fits-None? Improving Test Generation Using Context-Optimized Fitness Functions. *Proceedings of the 12th International Workshop on Search-Based Software Testing (SBST'19)*. Montreal, Canada, May 2018. Available at <http://greg4cr.github.io/pdf/19sbst.pdf>.
66. Gregory Gay. To Call, or Not to Call: Contrasting Direct and Indirect Branch Coverage in Test Generation. *Proceedings of the 11th International Workshop on Search-Based Software Testing (SBST'18)*. Gothenburg, Sweden, May 2018. Available at <http://greg4cr.github.io/pdf/18sbstdbc.pdf>.
67. Gregory Gay. Multifaceted Test Suite Generation Using Primary and Supporting Fitness Functions. *Proceedings of the 11th International Workshop on Search-Based Software Testing (SBST'18)*. Gothenburg, Sweden, May 2018. Available at <http://greg4cr.github.io/pdf/18sbstposition.pdf>.
68. Gregory Gay, Matt Staats, Michael Whalen, and Mats P.E. Heimdahl. Moving the Goalposts: Coverage Satisfaction is Not Enough. *Proceedings of the 7th International Workshop on Search-Based Software Testing (SBST'14)*. Hyderabad, India, June 2014. Available at <http://greg4cr.github.io/pdf/14sbst.pdf>.
69. Gregory Gay and Mats P.E. Heimdahl. Towards Community-Assisted Software Engineering Decision Making. *Proceedings of the 2nd International Workshop on Realizing Artificial Intelligence Synergies in Software Engineering (RAISE 2013)*. San Francisco, California, May 2013. Available at <http://greg4cr.github.io/pdf/13raise.pdf>.
70. Tim Menzies, Burak Turhan, Gregory Gay, Ayse Bener, Bojan Cukic and Yue Jiang. Implications of Ceiling Effects in Defect Predictors. *Proceedings of the 4th International Workshop on Predictive*

Models in Software Engineering (PROMISE'08). Leipzig, Germany, May 2008. Available at <http://greg4cr.github.io/pdf/08ceiling.pdf>.

Other Publications:

71. Mirosław Staron, Silvia Abrahao, Gregory Gay, Alexander Serebrenik. Testing, Debugging, and Log Analysis With Modern AI Tools. *IEEE Software (Practitioners' Digest)*. Volume 41, Issue 2. 2024. Available at <http://greg4cr.github.io/pdf/24digest.pdf>.
72. Paul Ralph, Nauman bin Ali, Sebastian Baltes, Domenico Bianculli, Jessica Diaz, Yvonne Dittrich, Neil Ernst, Michael Felderer, Robert Feldt, Antonio Filieri, Breno Bernard Nicolau de Frana, Carlo Alberto Furia, Gregory Gay, Nicolas Gold, Daniel Graziotin, Pinjia He, Rashina Hoda, Natalia Juristo, Barbara Kitchenham, Valentina Lenarduzzi, Jorge Martnez, Jorge Melegati, Daniel Mendez, Tim Menzies, Jefferson Moller, Dietmar Pfahl, Romain Robbes, Daniel Russo, Nytyi Saarinki, Federica Sarro, Davide Taibi, Janet Siegmund, Diomidis Spinellis, Mirosław Staron, Klaas Stol, Margaret-Anne Storey, Davide Taibi, Damian Tamburri, Marco Torchiano, Christoph Treude, Burak Turhan, Xiaofeng Wang, Sira Vegas. ACM SIGSOFT Empirical Standards for Software Engineering Research. *arXiv preprint arXiv:2010.03525*. March, 2021.
73. Yang Ren, Gregory Gay, Christian Kästner, and Pooyan Jamshidi. Understanding the Nature of System-Related Issues in Machine Learning Frameworks: An Exploratory Study. *arXiv preprint arXiv:2005.06091*. May, 2020.
74. Michael Unterkalmsteiner, Tingting Yu, Gregory Gay, Elizabeth Bjarnason, Markus Borg, Michael Felderer. Summary of the 5th International Workshop on Requirements Engineering and Testing (RET 2018). *ACM SIGSOFT Software Engineering Notes*. Volume 44, Number 1, March, 2019. Pages 31–34..
75. Markus Borg, Elizabeth Bjarnason, Michael Unterkalmsteiner, Tingting Yu, Gregory Gay, Michael Felderer. Summary of the 4th International Workshop on Requirements Engineering and Testing (RET 2017). *ACM SIGSOFT Software Engineering Notes*. Volume 42, Number 4. January, 2018. Pages 28–31. Available at <http://greg4cr.github.io/pdf/18ret.pdf>.
76. Michael Unterkalmsteiner, Gregory Gay, Michael Felderer, Elizabeth Bjarnason, Markus Borg, Mirko Morandini. Summary of the 3rd International Workshop on Requirements Engineering and Testing (RET 2016). *ACM SIGSOFT Software Engineering Notes*. Volume 41, Number 3. May, 2016. Pages 31–33. Available at <http://greg4cr.github.io/pdf/16ret.pdf>.
77. Gregory Gay, Giuliano Antoniol. 8th International Workshop on Search-based Software Testing (SBST 2015). *Proceedings of the 37th International Conference on Software Engineering (ICSE'15)—Workshop Summaries*. Florence, Italy, May 2015. Available at <http://greg4cr.github.io/pdf/sbst-summary.pdf>.
78. Gregory Gay. Automated Steering of Model-Based Test Oracles to Admit Real Program Behaviors. *Doctoral Dissertation, University of Minnesota*. Minneapolis, MN, May 2015. Available at <http://greg4cr.github.io/pdf/GregoryGayDissertation.pdf>.
79. Gregory Gay and Mats P.E. Heimdahl. Towards Community-Assisted Software Engineering Decision Making. *University of Minnesota Tech Report 13-015*. Minneapolis, MN, April 2013. Available at <http://greg4cr.github.io/pdf/13raise.pdf>.
80. Gregory Gay. The Robust Optimization of Non-Linear Requirements Models. *MS Thesis, West Virginia University*. Morgantown, WV, May 2010. Available at http://greg4cr.github.io/pdf/thesis_v1.pdf.

Invited Presentations:

1. Future Advanced Testing Technologies Workshop (FATTW). September 2025. Paris, France. Exploring the Integration of Large Language Models in Industrial Test Maintenance Processes

2. Mutation Workshop. April 2025. Naples, Italy.
Mutation Testing in the Trenches
3. AI in Healthcare Seminar. January 2025. Gothenburg, Sweden.
Testing Models and ML-Based Software
4. AI in Healthcare Seminar. January 2024. Gothenburg, Sweden.
Testing Models and ML-Based Software
5. International Conference on Software Testing (ICST). April 2023. Dublin, Ireland.
Invited Panelist: Doctoral Symposium
6. Dagstuhl Seminar 23103: Testing and Debugging of Data Analysis Workflows. March 2023. Wadern, Germany. Invited Participant
7. International Workshop on Artificial Intelligence in Software Testing (AIST). April 2021. Online.
Keynote: Learning How to Test - Generating Context-Infused Test Cases
8. International Conference on Software Engineering (ICSE). July 2020. Seoul, South Korea.
Invited Panelist: Student Mentoring Workshop
9. Chalmers University of Technology (Machine Learning Seminar). March 2020. Gothenburg, Sweden.
Invited Talk: Learning How to Search: Generating Exception-Triggering Tests Through Adaptive Fitness Function Selection
10. Jeppesen Systems AB. December 2019. Gothenburg, Sweden.
Invited Talk: An Introduction to Search-Based Test Generation
11. SAST Vst. October 2019. Gothenburg, Sweden.
Invited Talk: A Brief Introduction to Search-Based Test Generation
12. Shonan Seminar 160: Fuzzing and Symbolic Execution: Reflections, Challenges, and Opportunities. September 2019. Kanagawa, Japan.
Invited Talk: A Brief Introduction to (Metaheuristic) Search-Based Test Generation
13. South Carolina Law Review 2016 Symposium. February 2016. Columbia, SC.
Panelist: The Science of Cyber Attacks
14. University of Minnesota Graduate Student Colloquium. October 2011. Minneapolis, MN.
Invited Talk: Software Test Oracles: How I Learned to Stop Worrying and Love Mutation Testing
15. Midwest Verification Day 2011. September 2011. Minneapolis, MN.
Invited Talk: Towards Oracle Creation Support
16. Tsinghua University School of Software. March 2010. Beijing, PRC.
Invited Talk: Finding Robust Solutions to Model Optimization Problems
17. Institute of Software, Chinese Academy of Sciences. January 2010. Beijing, PRC.
Invited Talk: OURMINE: A Toolkit for Sharing Experiments
18. NASA Ames Research Center. August 2009. Mountain View, CA.
Invited Talk: Automatically finding the control variables for complex system behavior
19. WVU/NETL/ERA Workshop on Digital Preservation of Complex Engineering Data. April 2009. Morgantown, WV. Poster Presentation: Information Retrieval with HAMLET

Affiliate:

- Senior Member, IEEE.
- Member, ACM, Upsilon Pi Epsilon.